

PROCESS BOOK

THE STORY

SUSTAINABILITY IS NOT A SINGLE-VARIABLE PROBLEM

Sustainability is often reduced to simplified narratives: carbon emissions, renewable energy share, urbanization, or deforestation. While these indicators are useful individually, they rarely capture the complexity of how environmental systems interact.

Our objective is to lessen the knowledge gap through smart and interactive visualizations:

We explore the relationship between three major global systems:

- Energy production
- Deforestation
- Population density and urbanization

The objective is not to provide a single answer, but to help users visually investigate how these dimensions overlap, diverge, and influence one another.

MOTIVATION AND RESEARCH QUESTIONS

Climate change and biodiversity loss are usually discussed independently, despite being deeply interconnected.

Public opinion often assumes direct relationships:

- More fossil fuels → more deforestation
- Renewable energy → environmental protection
- Population growth → forest loss

However, when combining spatial datasets at a global scale, these relationships become much less straightforward.

TARGET AUDIENCE

MAKE YOUR AVERAGE JOE SUSTAINABLE

The visualization targets the general public and not experts. Environmental discussions are often inaccessible because they rely on technical terminology and scientific metrics.

Our objective was therefore to:

- Create an intuitive exploratory interface
- Encourage curiosity and interaction
- Preserve scientific accuracy while remaining approachable

THE DATA

DATASETS USED ([LINK](#))

Our project combines three datasets:

1. Global Power Plant Database (GPPD)

A CSV dataset containing:

- Power plant locations
- Installed capacity
- Primary fuel type

2. Tree Cover Loss

A GeoTIFF dataset describing:

- Deforestation locations
- Main deforestation drivers
- Land-use change categories

Categories include:

- Commodity-driven agriculture
- Shifting agriculture
- Forestry
- Urbanization
- Wildfires

3. Gridded Population

A GeoTIFF dataset representing:

- Population density distribution

PREPROCESSING

Both the population and deforestation datasets were initially stored as high-resolution GeoTIFF files containing hundreds of thousands of pixels.

Rendering these directly in the browser was not the best option :

- Less available tools
- Excessive memory consumption
- Slow rendering performance
- Large file sizes

To address this, we:

1. Converted raster data into GeoJSON points for the deforestation and polygons for population
2. Aggregated neighboring pixels
3. Downsampled the data spatially

This allowed us to find a sweetspot between precision of the data and the interactiveness/responsiveness.

WHAT WE OBSERVED ABOUT THE DATA

Deforestation cannot be explained solely through energy production.

Instead, different drivers emerge depending on the region:

- Commodity agriculture dominates in tropical regions
- Wildfires dominate in boreal forests
- Urbanization follows population growth
- Forestry activities cluster around industrial extraction zone

Population acts as indirect pressure, its growth correlates with:

- Urban expansion
- Food demand
- Infrastructure development
- Increased energy consumption

However, the relationship remains indirect and influenced by economics, geography, and political systems.

One of the most important observations was the absence of a linear correlation between fossil energy share and definitive deforestation within a country.

Countries with highly renewable grids can still experience severe forest loss, while some fossil-heavy countries maintain relatively stable forest coverage. This became one of the key narrative elements of the project.

DESIGN PROCESS

EXPLORATION FIRST

Our first design approach was intentionally broad. Before deciding on a final narrative, we explored as many visualizations as possible:

- Scatterplots
- Heatmaps
- Choropleths
- Histograms
- Linked plots
- Spatial overlays
- Treemaps
- Density maps

The objective was to discover what relationships were visually meaningful before keeping only relevant figures. This exploratory phase was essential to give impactful insights. After generating numerous visualizations, we progressively refined the project by asking:

- Does this visualization communicate something meaningful?
- Is the interaction intuitive?
- Does it support the narrative?
- Does it add insight or only visual complexity?

This process led us toward a smaller set of coordinated visualizations linked around a central interactive map.

DESIGN CHOICES

NEWSPAPER LAYOUT

One of our most important design decisions was adopting a newspaper-inspired visual identity. We wanted the website to feel like an interactive long-form article rather than a technical dashboard. We found it was an adequate format to support our visualizations and explanatory text.

The newspaper aesthetic supports:

- Narrative storytelling
- Structured reading flow
- Editorial credibility
- Visual immersion

This influenced:

- Typography choices
- Section hierarchy
- Pull quotes
- Overall layout
- Regional stories

Our final layout is thus made of a main title and introduction section with newspaper-like quotes. When scrolling down, three “case studies” present our narrative and allows a first interaction of the user with the plots. Below these stories the user is presented with our main plot box which is comprised of an interactive map and scrollable linked plots to the side of it. Finally an end section was added with conclusions regarding the presented data and static data driven plots.

VISUALIZATIONS

INTERACTIVE MAP

The map is the core component of the experience.

It allows users to:

- Navigate globally
- Zoom into regions
- Filter overlays
- Explore local patterns
- Trigger updates in linked visualizations

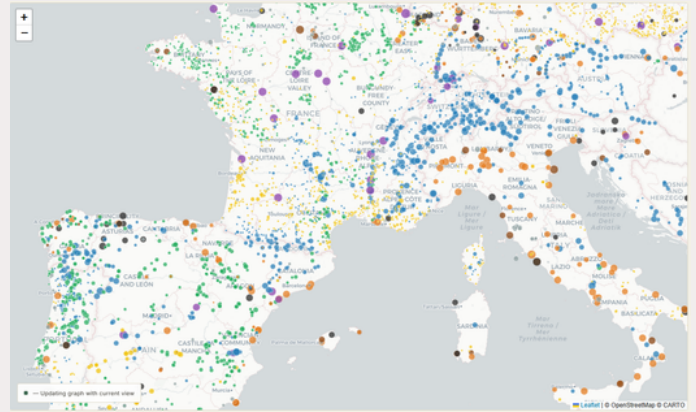
The map combines:

- Power plant markers
- Deforestation overlays
- Population layers

Users can dynamically compare how these systems overlap spatially.



Map filters



Interactive map

LINKED PLOTS

Several secondary visualizations update dynamically according to the current map view and which filters are selected. These plots contain but are not limited to:

Plant Statistics

Displays the number and distribution of visible power plants.

Capacity Treemap

Shows installed energy capacity by fuel type.

Fossil Share vs Forest Loss Scatterplot

Compares:

- Fossil energy share
- Definitive deforestation share
- Plant count

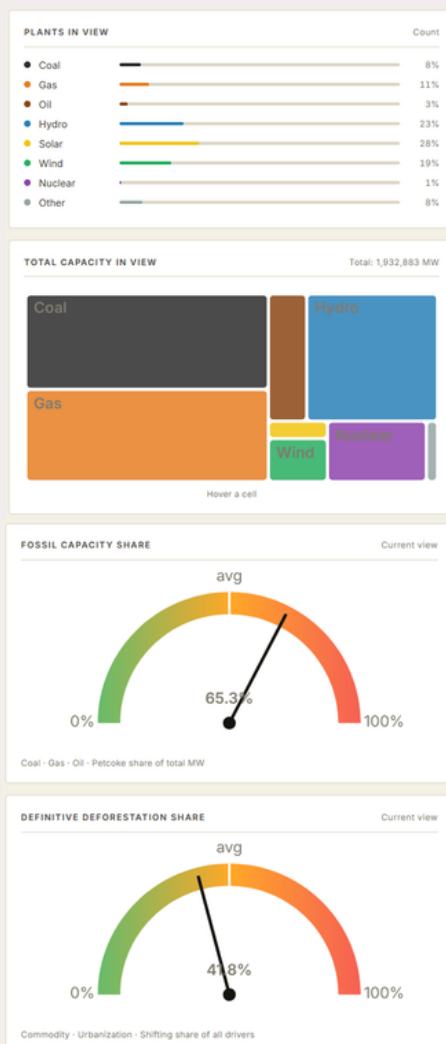
This visualization became central to our conclusion because it reveals the absence of a simple correlation.

Deforestation Share Charts

Displays the relative importance of different deforestation drivers.

Country Rankings

Highlights the most deforested countries currently visible in the viewport



Linked plots

VISUALIZATIONS

INTERACTIVE DATA STORIES

To make the experience more narrative-driven, we created three guided regional stories:

The Amazon Basin

Illustrates how a mostly renewable energy grid can coexist with severe commodity-driven deforestation.

Siberia in Flames

Shows the role of climate-driven wildfires in boreal forest loss.

Borneo for Business

Demonstrates how fossil extraction, and commodity agriculture interact within the same industrial system.

Three regions, three different stories

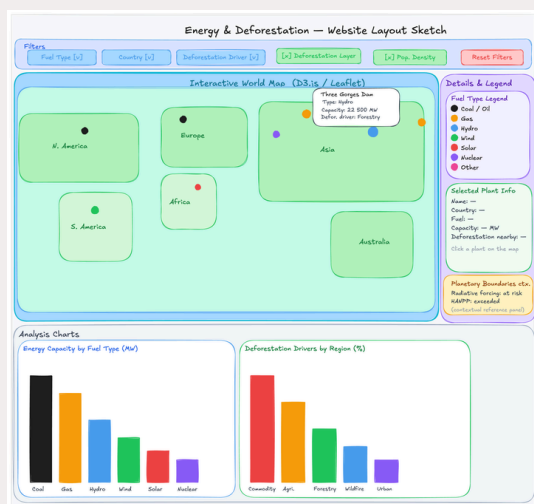


Interactive data stories

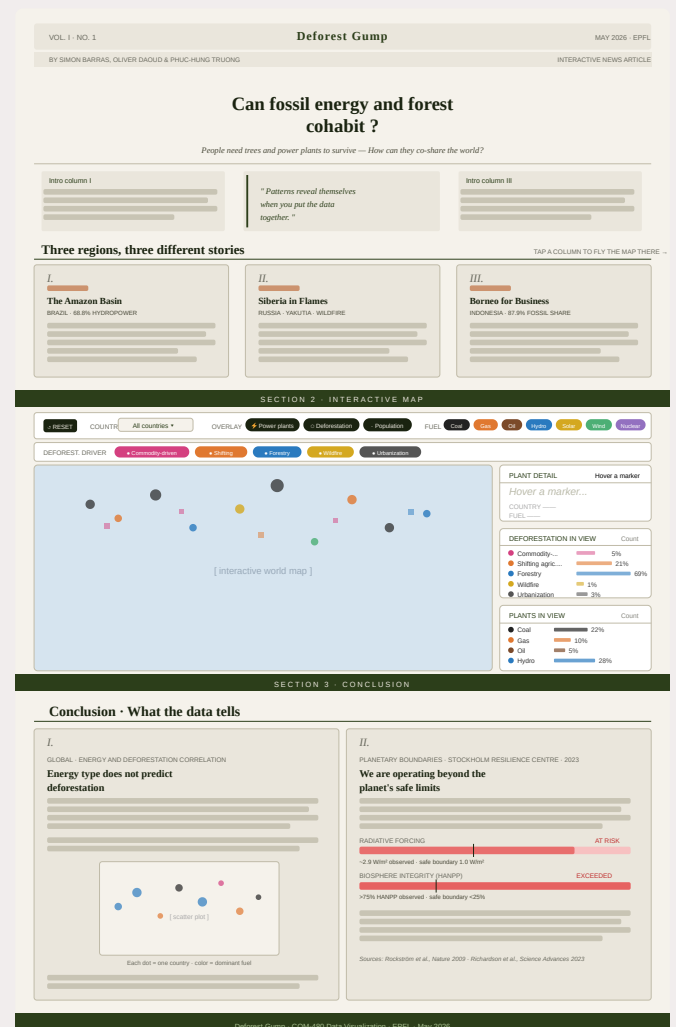
These dispatches act as entry points into the broader global visualization. The user can click on any of the stories and he will be redirected to the main plot with appropriate zoom and filters to view the story by himself

FINAL LAYOUT

Putting all these elements together finally give us the following layout. The user is guided through the story and at the same time can play with the visualizations to draw his own conclusions. The final layout stayed close to our original sketches with the addition of the newspaper theme and surrounding stories. One main change we orchestrated was putting the linked plots as a scrollable side bar next to the map instead of underneath. This allowed to view the plots and the map at the same time, giving a better user experience.



Initial layout sketch



Final layout sketch

CHALLENGES

MEMORY USAGE

The primary technical challenge was browser performance.

Large geospatial datasets quickly exceeded memory limitations, especially when rendering thousands of SVG or Leaflet elements simultaneously.

To solve this we:

- Downsampled raster datasets
- Reduced point density
- Simplified geometry
- Limited unnecessary redraws
- Aggregated statistics dynamically

Balancing detail and performance was a constant trade-off throughout development.

COMPREHENSIVENESS VS READABILITY

Another major challenge was deciding how much information to expose.

Environmental systems are inherently complex, but excessive detail can overwhelm users.

We therefore continuously balanced:

- Scientific precision
- Accessibility
- Visual clarity
- Narrative simplicity
- User engagement

This trade-off strongly influenced our final layout and interaction choice

WHAT WORKED

Several aspects of the project proved particularly successful:

- The newspaper narrative gave the project a distinctive identity
- The dispatch stories helped contextualize abstract data
- Linked visualizations encouraged exploration
- The map-centered interaction felt intuitive
- The conclusion section effectively synthesized the observations

Most importantly, the project succeeded in communicating that sustainability issues cannot be reduced to a single metric.

LIMITATIONS

Despite the final result, several limitations remain.

Data Resolution

Downsampling was necessary for performance but reduced geographic precision.

Correlation vs Causation

The visualization highlights spatial relationships but cannot prove causality.

Global Generalization

Country-level aggregation sometimes hides important local dynamics.

Browser Constraints

Rendering large-scale geospatial information in real time remains computationally expensive.

PEER ASSESSMENT

PHUC-HUNG TRUONG - CGO

The Chief Graphical Officier served as the visual backbone of the project. He introduced the three data story articles and implemented their interactive integration with the map. Throughout development, he maintained visual coherence across the entire interface, iteratively refining the graphical style, standardizing color palettes, and ensuring a consistent user experience.

OLIVER DAOUD - CSO

Our Chief Sustainability Officier laid the conceptual foundation of the project by sourcing the core dataset and proposing the initial idea. He took full ownership of the planetary boundaries narrative, developing the data stories around ecological limits and their global implications. He also designed and implemented the dynamic chart that brings those trends to life.

SIMON BARRAS - CCO

Simon, the Chief Chart Officier was the primary engineer behind the website's data visualizations. He built and refined the majority of the charts, optimized the rendering and display of map overlays for performance and clarity, and extended the interface with additional graphs and interactive filters to enrich the user's exploratory experience.